

9. (a) Give the observations made during the reactions of lithium, sodium and potassium with water. Explain the trend in reactivity in terms of electronic structure. [6 QER]

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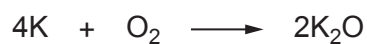
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- (b) Potassium reacts with oxygen to produce potassium oxide.



- (i) Calculate the maximum mass of potassium oxide that could be produced when 15.6 g of potassium reacts completely with oxygen. [3]

$$A_r(\text{K}) = 39$$

$$A_r(\text{O}) = 16$$

Mass = g

- (ii) Another reaction requires 0.050 mol of oxygen gas. Calculate the number of molecules in 0.050 mol of oxygen gas. Give your answer in **standard form**. [2]

$$\text{Avogadro's number} = 6.0 \times 10^{23}$$

Number of molecules =

